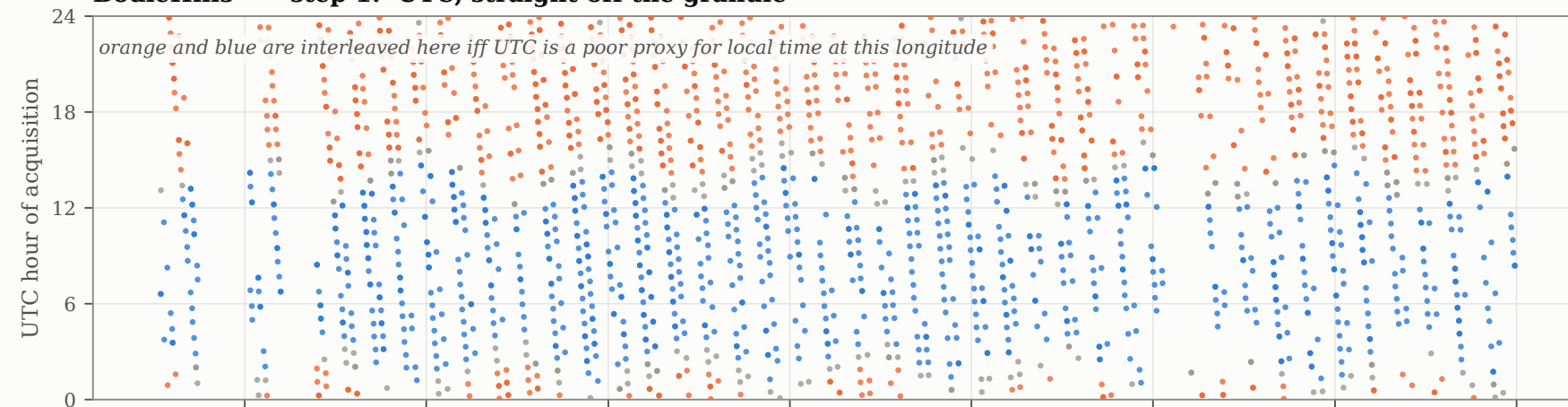
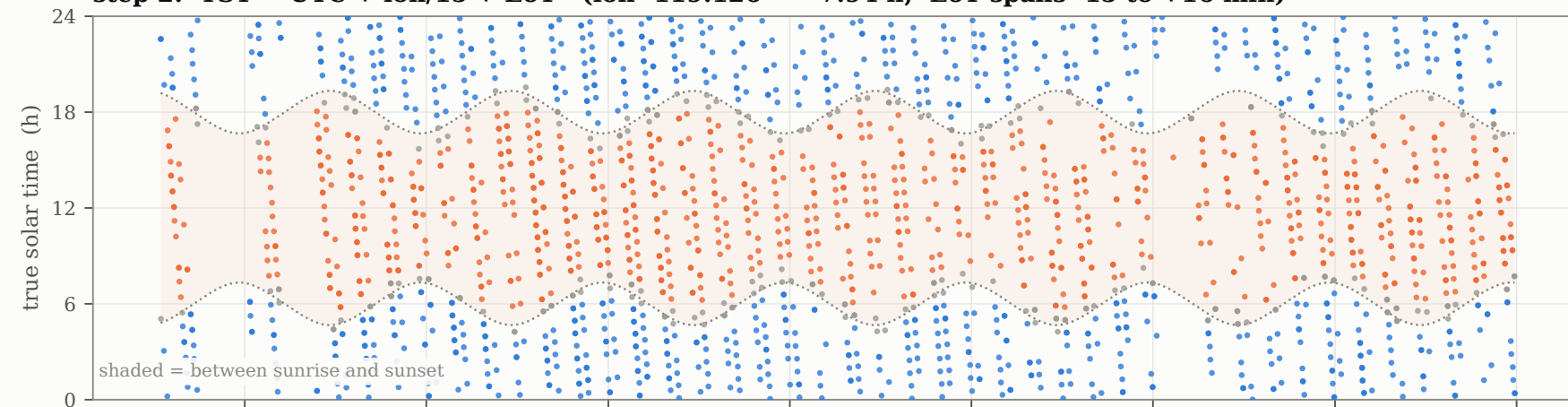


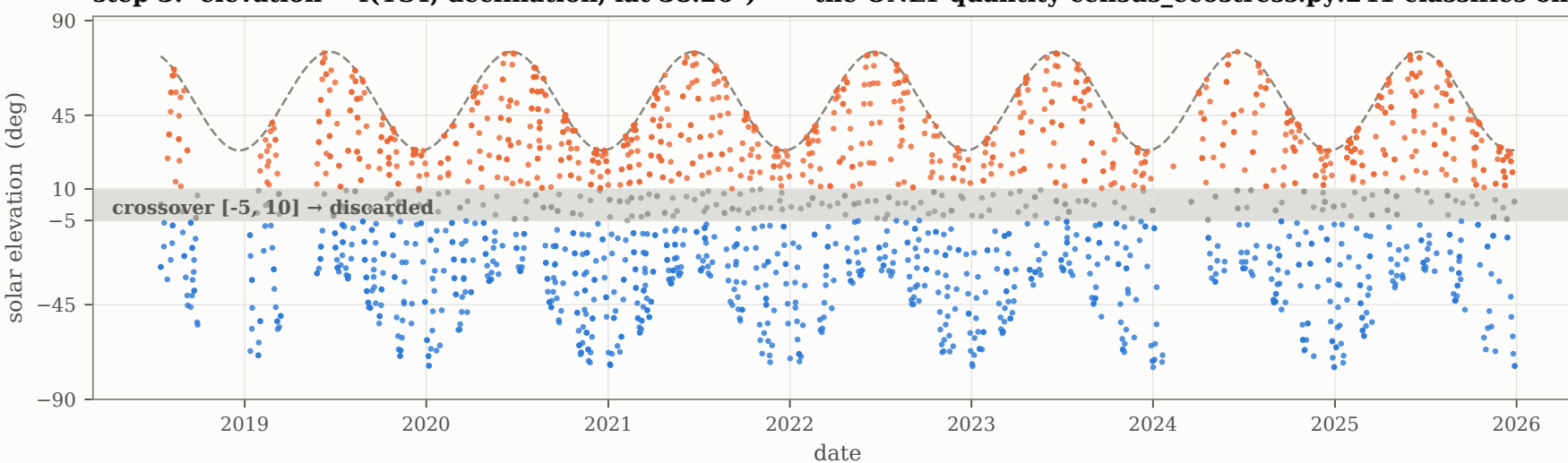
BodieHills — step 1: UTC, straight off the granule



step 2: TST = UTC + lon/15 + EoT (lon -119.126° → -7.94 h; EoT spans -15 to +16 min)



step 3: elevation = f(TST, declination, lat 38.26°) — the ONLY quantity census_ecostress.py:241 classifies on



worked examples

UTC	TST	elev	phase
2018-07-16T06:36	22.57	-27.1	night
2019-08-14T17:51	9.85	52.7	day
2020-03-05T08:46	0.63	-57.0	night
2020-09-27T23:31	15.76	23.6	day
2021-03-04T15:41	7.55	13.3	day
2021-09-07T07:09	23.26	-45.1	night
2022-05-07T07:28	23.59	-34.7	night
2023-01-17T09:23	1.28	-66.0	night
2023-09-15T03:18	19.47	-15.7	night
2024-09-22T23:15	15.46	28.2	day
2025-04-30T15:10	7.28	23.9	day
2025-12-28T08:23	0.41	-74.1	night

longitude offset -7.94 h
median |TST - UTC| 8.01 h

UTC cannot be read as local time here: the same UTC hour lands in daylight or in darkness depending on the date.